

QoE Characterization for Video-on-Demand Services in 4G WiMAX Networks

Amitabha Ghosh, Malolan Chetlur, Partha Dutta, Umamaheshwari Devi, Shivkumar Kalyanaraman
amitabha.ghosh@gmail.com, Next Generation Telecom Research, IBM India Research Laboratory, Bangalore

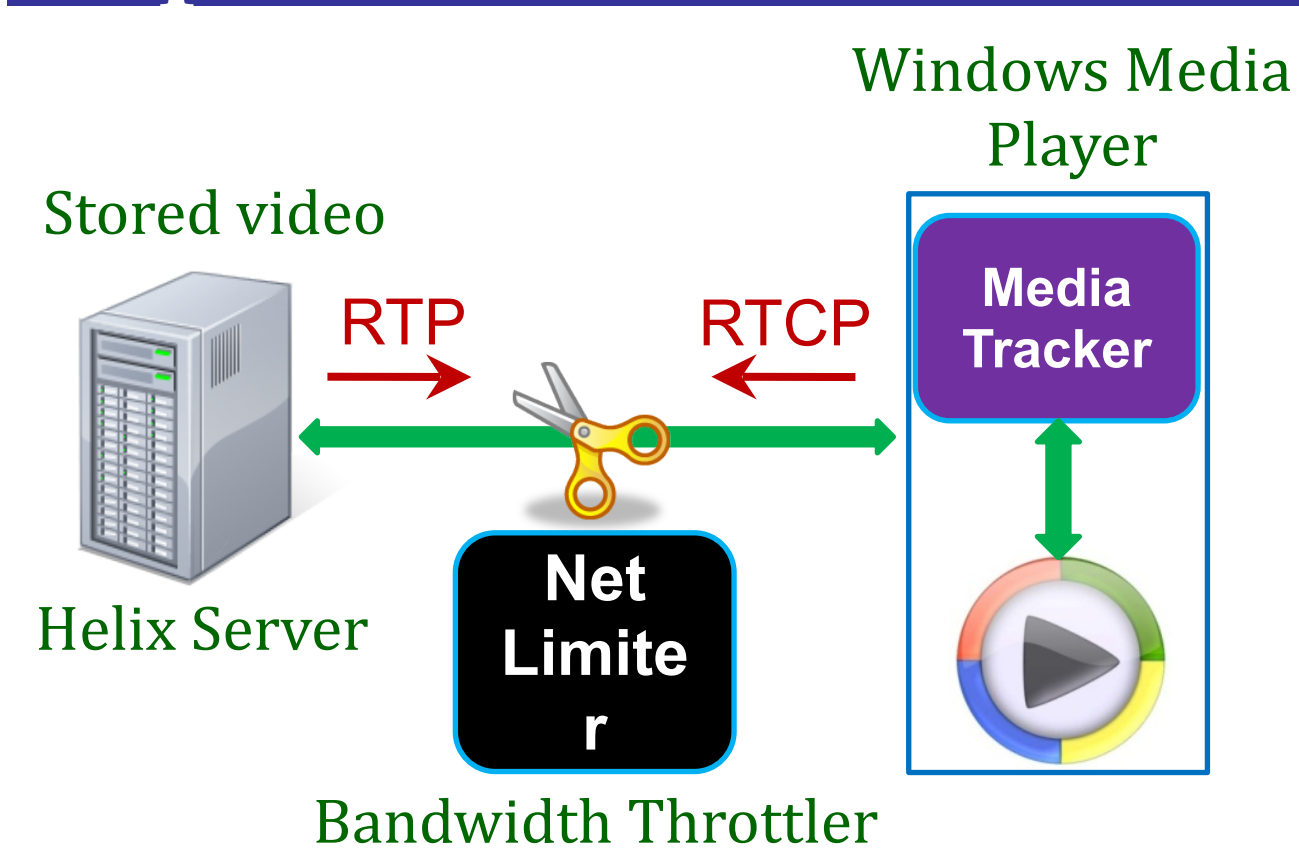
Motivation

- ❖ Active deployment of **triple** and **quadruple services** by network operators delivering voice, real-time high-quality video, and value-added data services over converged networks (e.g., 4G).
- ❖ Need to guarantee subscribers' **Quality-of-Experience (QoE)** and provide differentiated services in the face of heterogeneous end devices, varying wireless channels, resource constraints, etc.
- ❖ Lack of **QoE orchestration** models that map network events (e.g., variation in bandwidth, delay, error rates, jitter) to subscribers' QoE.

Problem Statement

- ❖ To build robust QoE models by measurement and detection of streaming **Video-on-Demand (VoD)** characteristics and network parameters over 4G WiMAX networks.
- ❖ To propose orchestration mechanisms using the above QoE model

Experimental Evaluation

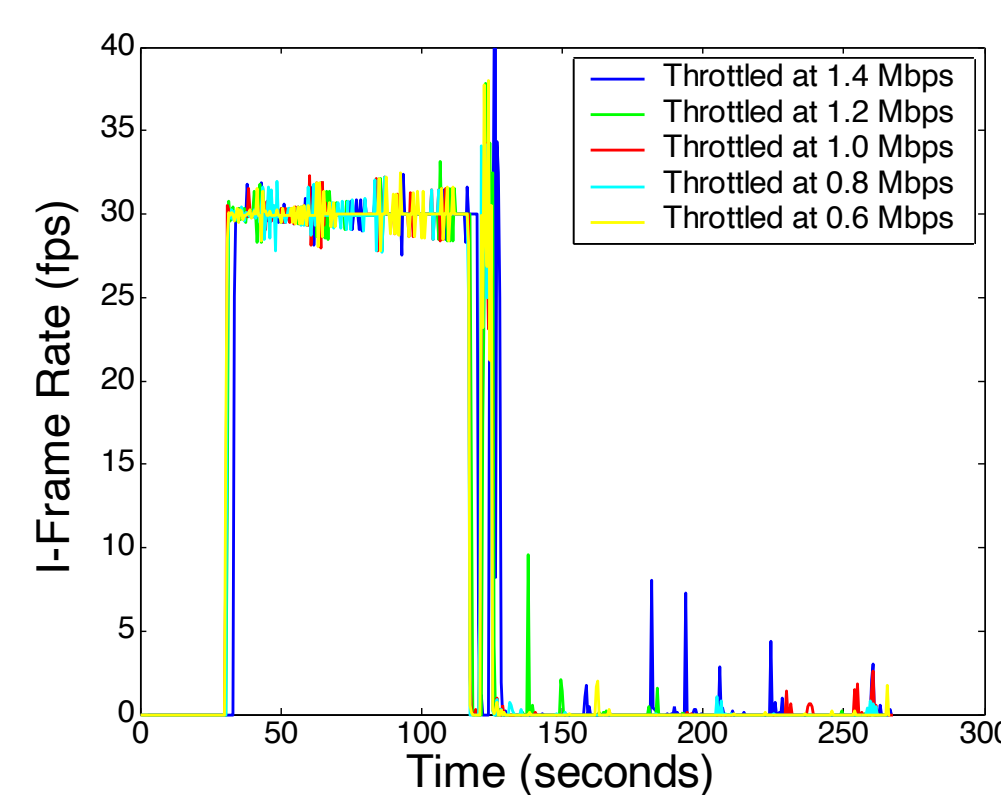
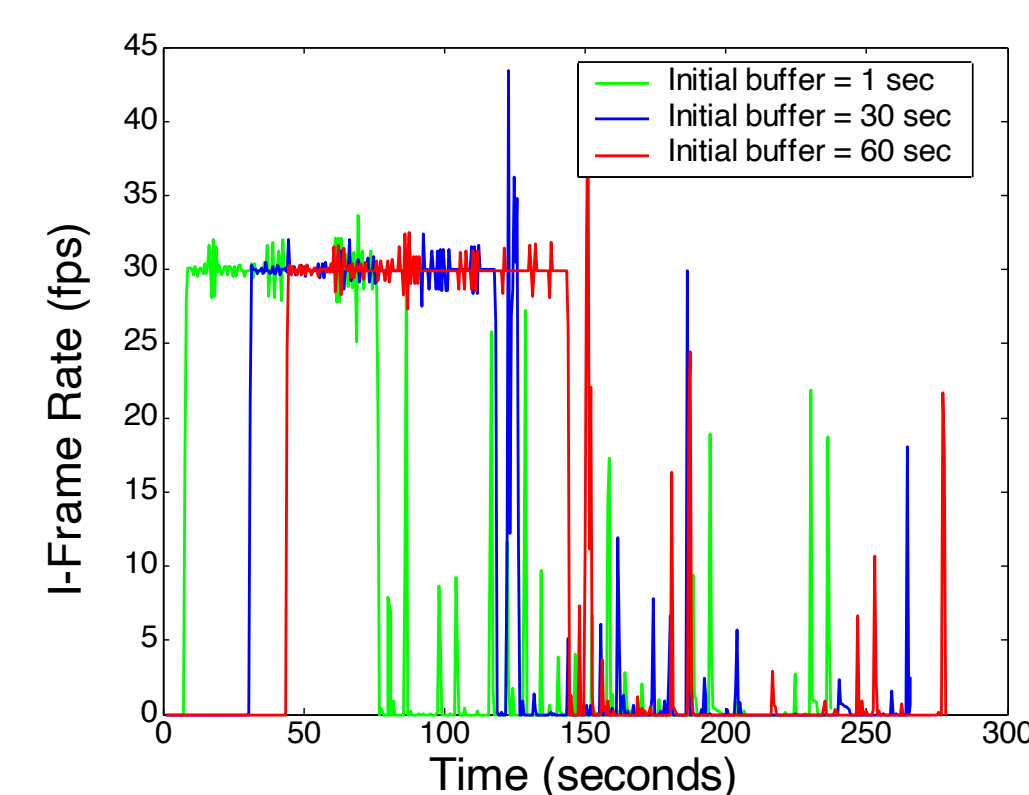


Video Characteristics

- ❖ Bit-rate: 1.96 Mbps
- ❖ Duration: 230 secs
- ❖ File size: 56.7 MB
- ❖ Media Player initial buffering: 30 secs

Objective

- ❖ Measure streaming VoD characteristics under different network conditions (varying bandwidth, packet drops)



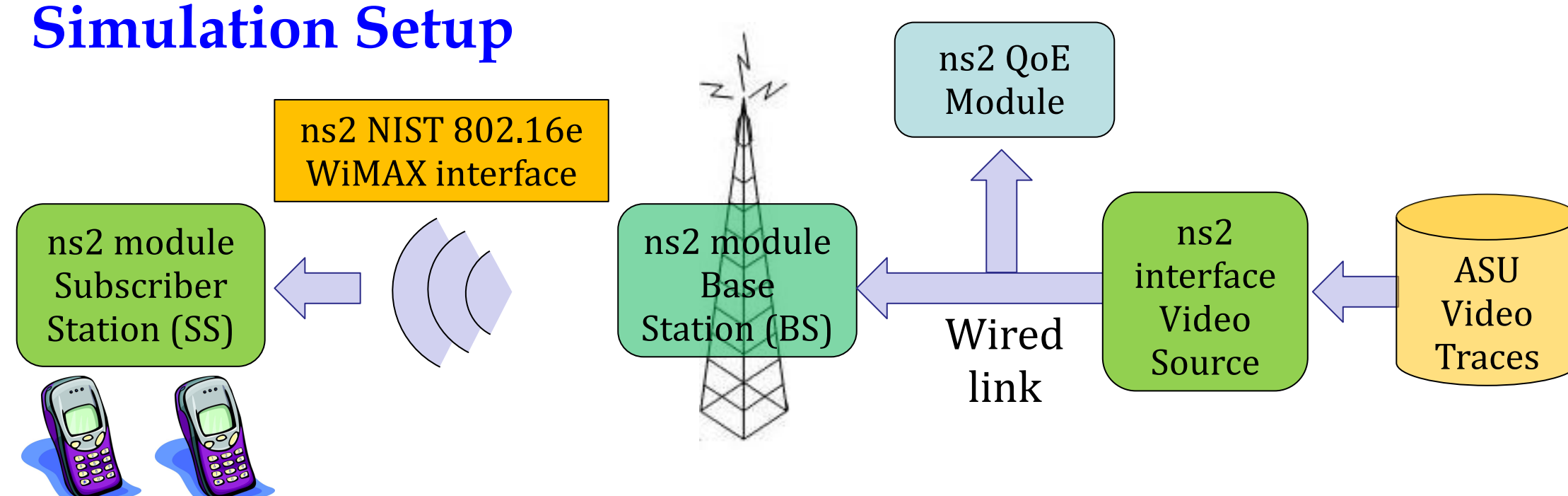
Variation of I-frame rate with different initial buffering and throttling bandwidths

On Going and Proposed Work

ns2 Simulator with WiMAX Module

- ❖ Feed publicly available video traces from ASU through an ns2 interface to simulate CBR/VBR VoD traffic
- ❖ Measure **VoD characteristics** (bit-rate, bandwidth, frame types, etc) using a ns2 QoE module
- ❖ Study the **variation of I-frame rate** at the SS with varying bandwidth, delay, packet error rate, and jitter
- ❖ Based on wireless channel characteristics, **set QoS parameters** in the BS to support differentiated QoE at SS

Simulation Setup

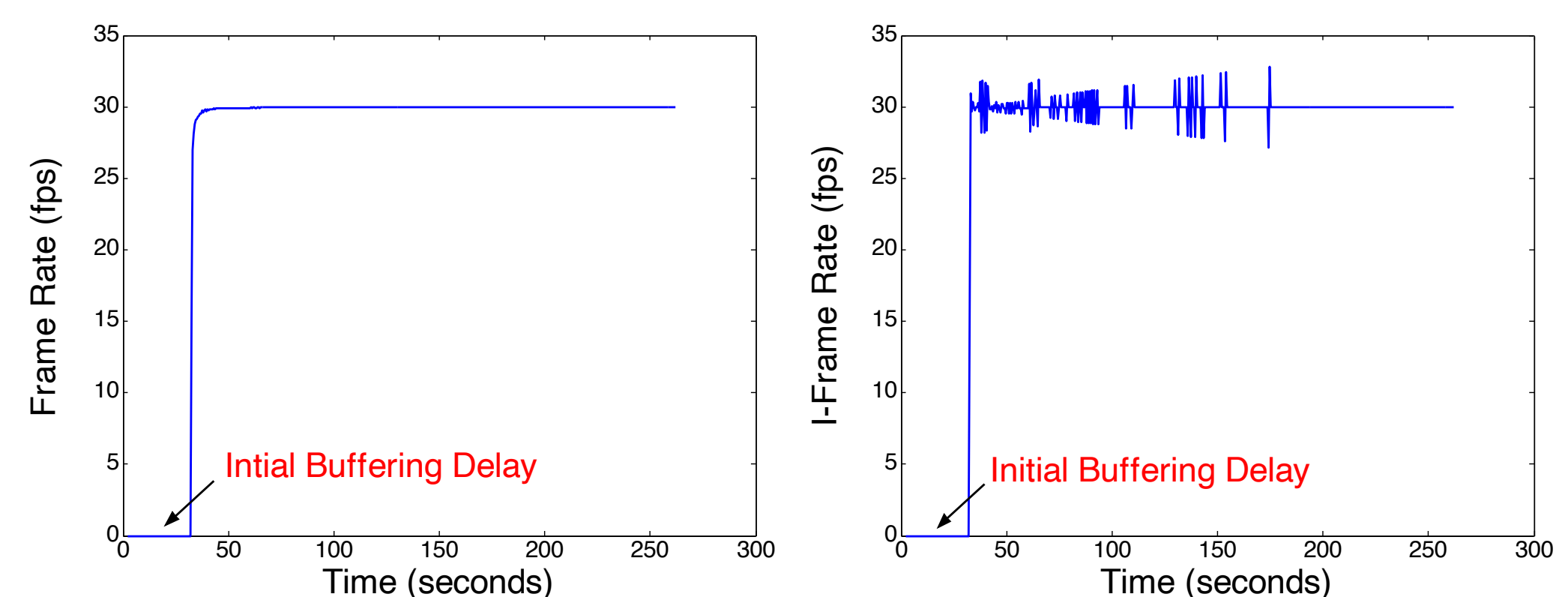


Business Context

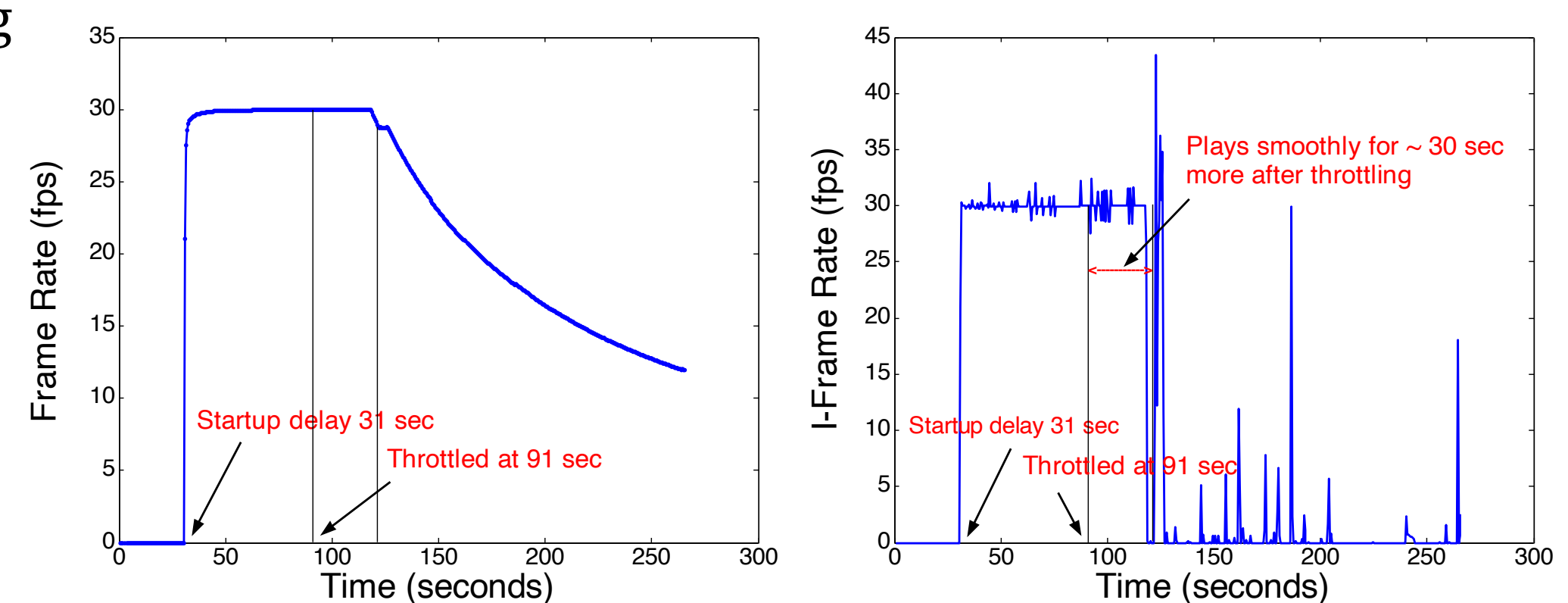
- ❖ Extending IBM's business model by using **PRISM**-based appliances (e.g., **Services Substrate Appliance – SSA**) into key infrastructure elements at various entry points (e.g., gateways, base-stations)

State-of-the-Art in QoE

- ❖ Session quality (initial buffering, re-buffering, audio video synchronization), video quality (frame quality, frame-rate, stalling), and nature of the content contribute to user' QoE.
- ❖ Several objective (e.g., **PSNR**, **SSIM**) and subjective metrics (e.g., **MOS**, **VQM**) to measure QoE – but unreliable.
- ❖ Bit-rate switching, feedback-based encoding, etc.
- ❖ Very little understanding of the major causes of user perceived degradation.



Without bandwidth throttling (no Net Limiter)



Bandwidth throttled to 1.6 Mbps at 60 sec from the start of the video

Observations

- ❖ Smooth video playback for initial buffering time after stalling
- ❖ Stalls frequently after 91 seconds, closely mimicking I-frame rate, irrespective of throttled bandwidth

Inferences

- ❖ I-frame rate is a better indicator of users' QoE than frame rate
- ❖ Small bandwidth variation and packet losses cause major degradation in viewers' QoE

References

- ❖ Mukundan V, S. Sengupta, M. Chatterjee, and R. Neogi, **Towards a Video QoE Definition in Converged Networks**, ICDC '07.
- ❖ F. H. P. Fitzek, P. Seeling, and M. Reisslein, **Using Network Simulators with Video Traces**, ASU, EE, Technical Report, 2003.
- ❖ M. Reisslein, et al., **Traffic and Quality Characterization of Scalable Encoded Video: A Large-Scale Trace-Based Study**, ASU, EE, Technical Report, 2003.